Moka Rochishnu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.85
Sri Chaitanya Junior College	Class XII (BIEAP)	2018	96.90%
Sri Chaitanya	Class X (BSEAP)	2016	10.00%

SKILLS

- **Hardware description Language**:Verilog
- **Programming**:C
- **Embedded software & MCUs:** Vitis IDE,CCS,TI C2000 (TMS320F28379D,77D)
- **Simulation Tools**: Cadence Virtuoso, Vivado, LTspice, MATLAB, PLECS, Proteus 8
- **EDA tools:** CAM365

MTECH THESIS

Guide: **Prof. Aneesh Shobhanan**

Title:Blind Photonic Polarization Demultiplexing of Dual-Polarization QPSK Using a Programmable PIC

Description:

Design and simulation of coherent optical polarization recovery system for dual polarization communication links by implementing SPSA technique, with the goal of replacing computationally intensive DSP using an integrated photonic MZI architecture.

RELEVANT COURSES

- **Analog Domain**: Analog IC Design and Analog IC Design LAB.
- **Digital Domain**: Digital IC Design**,** Introduction to VLSI Design, Digital IC Design Lab.
- **Memory course** : Memory device technology For AI/ML.
- **Microfabrication courses**: Introduction to MEMS, Nanoelectronics: Principles and Devices.

PROJECTS

LDO

- Designed a Low Drop-Out (LDO) regulator in Cadence Virtuoso using a **PMOS two-stage OTA error amplifier** coupled to a PMOS pass transistor to maximize input common-mode headroom under low DC supply constraints.
- Performed multi-corner **PVT variations** across TT, FF, and SS corners to validate sub-microsecond regulation responses under worst-case hardware stress

Two Stage Folded Cascode

- Designed and simulated a high-gain **NMOS Folded Cascode Amplifier** in Cadence Virtuoso, optimizing structural transistor sizing and bias conditions across multiple differential and common-source stages.

Counters

- Designed and simulated a 8 bit synchronous and asynchronous counters with Static and Dynamic architecture(Clock-Skew Insensitive Approach,TSPCR)

FPGA Implementation of GELU:

- Deployed a high-performance GELU activation function accelerator on a **Zynq UltraScale+ MPSoC FPGA** using a 16-bit Q8.8 fixed-point architecture.
- Packaged custom Verilog RTL modules into a memory-mapped **AXI4-Lite IP block (comm_0)**, interfacing

Programmable Logic (PL) seamlessly with the ARM Processing System (PS) via an AXI Interconnect.
- Developed low-level bare-metal driver applications in **C using Vitis IDE** (Xil_In32/Xil_Out32) to pass real-time test vectors across the PS-PL boundary and validate system throughput over UART.

WORK EXPERIENCE

- Worked as R&D Engineer (Power electronics) in Medha Servo Drives Pvt Ltd (Jul 2022-Jun 2023)
- Worked as Assistant Manager in Bharath Dynamics Limited (Aug 2023- Jul 2025)